

SOLUTIONING INTERNSHIP



AEROSPACE
ENGINEERING

Avotron Pte. Ltd.



PROJECT TITLE & SYNOPSIS:

Aircraft Documentation Platform Using Generative AI for Maintenance Manual Access

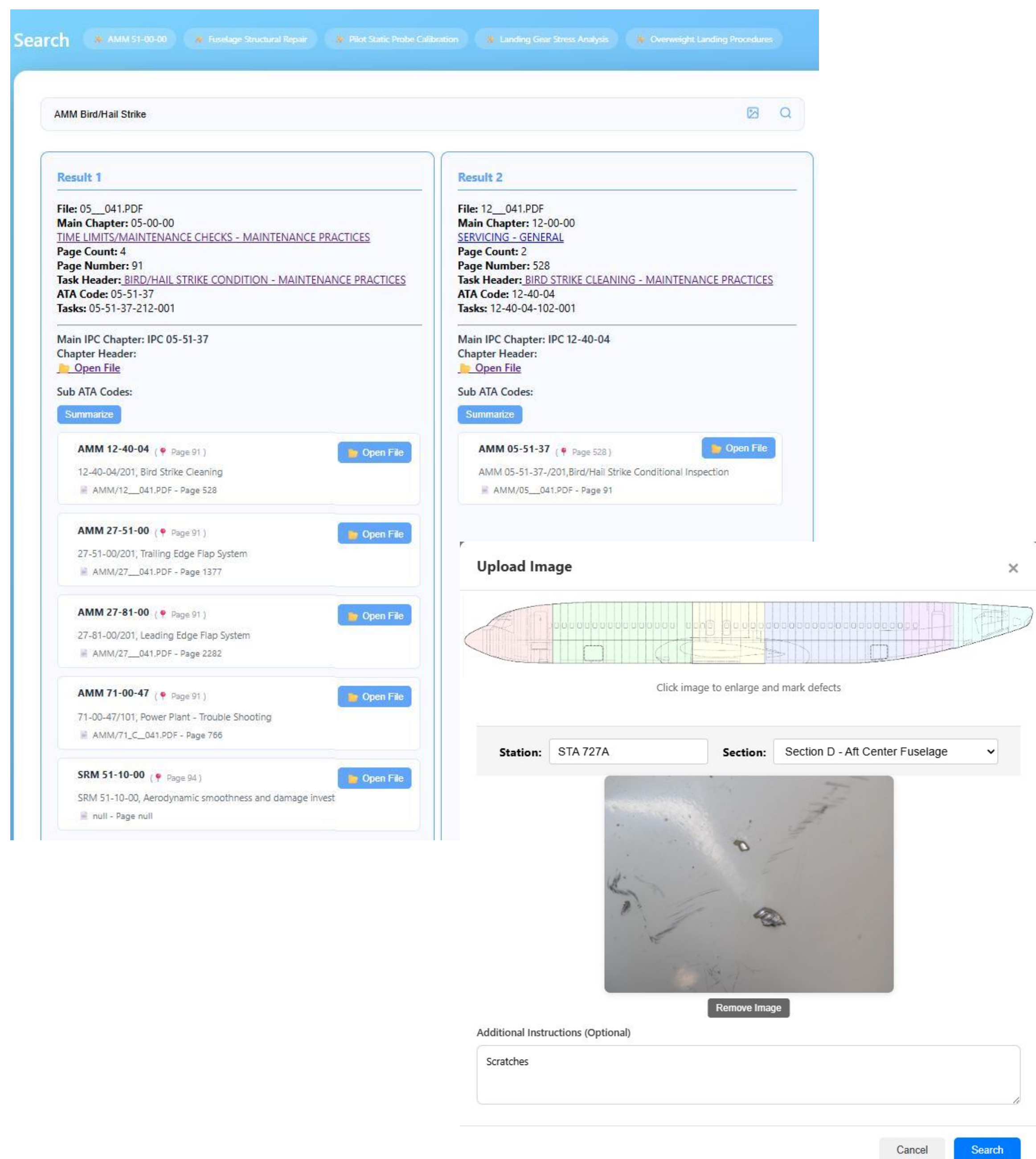
A web-based dashboard for aircraft maintenance specialists to efficiently locate relevant sections of technical manuals based on system or component queries leveraging on Generative AI technologies.

KEY AREAS OF RESPONSIBILITIES :

- Collaborate with clients to gather requirements and ensure solutions are accurately verified and validated.
- Design and implement scalable system architectures for dashboards to support growth and performance needs.
- Explore datasets and design preprocessing pipelines to ensure clean, structured, and reliable inputs for analysis.
- Research and development of Gen Ai (e.g. LLM, VLM) to understand user context and achieve accurate retrieval.
- Develop intelligent documentation search features by tagging systems and components, linking to ATA chapters, and enabling queries by part number, fault code, or system name.

IMPACT & OUTCOME:

- Fast access to the exact manual and relevant sections, reducing document search time from ~5min to ~30sec, allowing maintenance teams to be more productive.
- Improved retrieval accuracy by combining domain-specific chunking, clean preprocessing, and high-quality semantic embeddings.



COMPANY PROFILE:

- Avotron Pte. Ltd. manufactures and repair engineering and scientific instruments with wholesale trade of a variety of goods without a dominant product as the secondary activity.

PROBLEM STATEMENT:

- Aircraft maintenance specialists face inefficiencies in locating relevant information within vast technical manuals, leading to delays, increased workload, and potential risks to operational safety.

Student : Er Zhong Xun Ervin Class : DCPE/FT/3B/04
Company : Avotron Pte. Ltd.
Supervisor : Mr Ng Yong Thiang
LO : Mr Andrew Tan



SCHOOL OF
ELECTRICAL &
ELECTRONIC ENGINEERING